

Decoupling Stem-Cell Maintenance from Cellular Senescence: A Comparative Investigation of Meristem Longevity & Quality Control in *Ginkgo biloba*

Principal Investigator: Dr. Gemini Notebook Consortium

Target Organism: *Ginkgo biloba* L. (Vascular Cambium)

Duration: 3 Years (Months 1–36)

Total Direct Costs: \$1,000,000 | **Total Indirects:** \$250,000

Key Mechanisms: WOX4-PXY Axis, Ku70/80 DSB Repair, ATG8 Autophagy

Translational Model: Human Stem Cell Exhaustion & Aging

1. EXECUTIVE SUMMARY & SPECIFIC AIMS

Multi-century perennials like *Ginkgo biloba* possess the remarkable capacity to maintain active stem-cell populations and produce viable secondary tissues across hundreds or thousands of years without undergoing programmed cellular senescence. This project investigates whether this long-term survival is achieved through **heterochronic shielding** (actively maintaining juvenile developmental competence in lateral meristems) or through a robust network of **cellular quality control mechanisms** (structural double-strand DNA repair, autophagic clearance, and spindle checkpoint priming). By combining single-nuclei RNA sequencing (snRNA-seq), quantitative microRNA profiling, and genotoxic stress challenges, this proposal establishes a testable framework connecting plant stem-cell maintenance with human cellular resilience.

CORE RESEARCH AIMS:

- **Aim 1: Single-Nuclei Transcriptomic Mapping (snRNA-seq)** across 15y, 200y, and 600y+ *G. biloba* cambial layers using a specialized polyphenol-neutralizing extraction protocol (Tree-NEB).
- **Aim 2: Functional & Heterochronic Evaluation** of the *gbi-miR172–GbAP2–GbAIL1* and *GbWOX4–GbPXY* signaling axes across age cohorts.
- **Aim 3: Stress-Challenge Kinetics & Quality Control** measuring γ -H2AX double-strand DNA break resolution (Ku70/80 scaffold) and Bafilomycin A1-inhibited autophagic flux (ATG8-II/NBR1).

2. EXPERIMENTAL METHODOLOGY & WORKFLOW

Aim 1 Method: Tree-NEB snRNA-seq Protocol. To overcome the high tannin, polyphenol, and lignin barriers in ancient tree bark, inner bark/cambium cores will be cryo-ground and homogenized in specialized Tree-NEB buffer containing 2.0% PVP-40, 1.0% PEG-8000, 1.0% β -mercaptoethanol, and 0.1% Triton X-100. Intact nuclei will be purified over a 30%/40% OptiPrep density gradient, tagged with cholesterol-modified oligonucleotide (CMO) hashtags, and captured on a 10x Genomics Chromium platform targeting 15,000 single nuclei per age cohort.

Aim 2 Method: Heterochronic Axis Alignment & Single-Cell Pseudotime. Our bioinformatic alignments confirm **100.0% sequence identity** across the 60-aa WOX4 homeodomain and **98.2% conservation** across the dual AP2

domains of GbAIL1 between *Populus* and *Ginkgo*. Single-cell transcriptomes will be evaluated via Diffusion Pseudotime in Scanpy to test if 600y+ cambial stem cells retain juvenile *GbAIL1* expression matching 15y saplings.

Aim 3 Method: Genotoxic/Proteotoxic Stress Kinetics. Explants will be challenged with 10 μ M zeocin (DSB induction) or 0.5 mM H₂O₂ (proteotoxic stress). Double-strand break repair kinetics will be measured via γ -H2AX foci resolution half-life ($t_{1/2}$), and autophagic flux will be quantified via western blot tracking ATG8-I to ATG8-II conversion in the presence vs. absence of Bafilomycin A1 (100 nM).

3. 3-YEAR MILESTONE TIMELINE & DELIVERABLES

The research plan is structured across three synchronized phases, moving from field sampling and single-nuclei sequencing to functional stress challenges and translational human cell-line benchmarks.

Phase / Period	Milestone Task / Activity	Key Readout & Deliverable	Success Benchmark
Year 1 (M1–M6)	Field Core Sampling & Tree-NEB Protocol Optimization	High-purity nuclei isolated from 15y, 200y, and 600y+ <i>G. biloba</i> cambium	Nuclei yield >1,000/μL; RIN > 8.0; zero polyphenol carryover
Year 1 (M7–M12)	10x Chromium snRNA-seq & CMO Hashtag Sequencing	Single-nuclei cell atlas of vascular cambium across age cohorts	15,000 nuclei per age group; clear <i>GbWOX4</i> + cluster resolution
Year 2 (M13–M18)	Stem-Loop RT-qPCR & Heterochronic Ratio Mapping	Validation of <i>gbi-miR156</i> , <i>gbi-miR172</i> , and <i>GbAIL1</i> trajectories	Inversion point identified in pseudotime along cambial differentiation
Year 2 (M19–M24)	Acute Stress Challenges (Zeocin / H ₂ O ₂ / BafA1)	γ-H2AX foci resolution kinetics & ATG8 autophagic flux ratios	Quantified t _{1/2} repair half-life; validated autophagic flux
Year 3 (M25–M30)	Heterologous Expression of <i>GbKu70</i> in Human Cells	Transfection of <i>Ginkgo</i> Ku70 into senescent human cell assays	Evaluation of DSB repair rescue and senescence suppression
Year 3 (M31–M36)	Data Integration, Manuscript Preparation & Reporting	Final Research Report, Open-Access Codebase & Datasets	2 High-impact publications; public snRNA-seq database release

4. ITEMIZED 3-YEAR RESEARCH BUDGET

Budget allocation prioritizes high-throughput single-nuclei sequencing, specialized reagents for polyphenol extraction, and genotoxic stress kinetics.

Budget Category	Year 1 (\$)	Year 2 (\$)	Year 3 (\$)	Total Directs (\$)	Justification & Allocation
A. Personnel (PI, Postdoc, Bioinformatician)	140,000	145,000	150,000	435,000	Postdoc (1.0 FTE), Bioinformatician (0.5 FTE), PI (1 mo)
B. Sequencing & Genomics (10x / NovaSeq)	120,000	40,000	20,000	180,000	10x Chromium capture, CMO tags, NovaSeq 6000 runs
C. Reagents & Assays (Zeocin, BafA1, Ab)	35,000	45,000	30,000	110,000	OptiPrep, Antibodies, TaqMan probes, Zeocin, BafA1
D. Field Logistics & Core Sampling	25,000	15,000	10,000	50,000	Field trips to Guizhou/Zhejiang, core sampling permits

Budget Category	Year 1 (\$)	Year 2 (\$)	Year 3 (\$)	Total Directs (\$)	Justification & Allocation
E. Human Cell Line Assays & Culture	10,000	25,000	30,000	65,000	WI-38/IMR-90 cell lines, transfection reagents, assays
F. Publication & Open Data Access	5,000	10,000	15,000	30,000	Open-access publishing fees, repository hosting
TOTAL DIRECT COSTS	335,000	280,000	255,000	870,000	Direct Research Operations
INDIRECT COSTS (15% Overhead)	50,250	42,000	38,250	130,500	Institutional Facilities & Admin
GRAND TOTAL REQUESTED	385,250	322,000	293,250	1,005,500	Total Funding Requirement

5. TRANSLATIONAL IMPACT ON HUMAN CELLULAR AGING

By comparing the 100% conserved DNA-binding channels of *Ginkgo* Ku70 with human orthologs and evaluating how multi-century cambial stem cells decouple environmental stress from permanent senescent arrest, this project establishes a direct cross-kingdom bridge. The outcomes will inform novel therapeutic strategies to prevent stem cell exhaustion and preserve double-strand break repair fidelity in aging human tissues.